HYDRO CARBON (ALKANE + ALKENE + ALKYNE)

1. C —C and C —H bond length of C_2H_6 molecule is–
(1) 1.54 A° & 1.11 A° respectively
(2) 1.34 A° & 1.10 A° respectively
(3) 1.20 A° & 1.08 A° respectively
(4) 1.54 A° & 1.08 A° respectively

2. How many structures of C_6H_{14} are possible–
(1) 3 (2) 4
(3) 5 (4) 6

3. Which one has 4° carbon
(1) n-Pentane (2) Iso pentane
(3) Neo pentane (4) Iso hexane

4. IUPAC Name of $(CH_3)_2C(C_2H_5)_2$ is
(1) 3,3-Dimethyl pentane
(2) 1,1-Diethyl-1,1-di methyl methane
(3) 3- Ethyl-2-methyl pentane
(4) 3-Iso propyl pentane

5. Which of the following alkane can not be prepared by hydrogenation of olefine–
(1) Propane (2) Iso butane
(3) Iso pentane (4) Neo pentane

6. $R—CH=CH_2 + H_2 \xrightarrow{'x'} R—CH_2—CH_3$
Catalyst 'x' in above reaction which complete the reaction at lower temperature is–
(1) Ni (2) Pt
(3) Pd (4) Both (2) & (3)

7. From how many alkyl halide we get Iso pentane on reaction with Zn + HCl (only structural isomers)
(1) 2 (2) 3
(3) 4 (4) 5

8. $C_2H_5Br + Na + CH_3—CH(CH_3)—Br \xrightarrow{\text{dry ether}}$
Which is not product of above reaction :
(1) $CH_3—CH_2—CH_2—CH_3$
(2) $CH_3—CH_2—CH(CH_3)—CH_3$
(3) $CH_3—CH(CH_3)—CH(CH_3)—CH_3$
(4) $CH_3—CH_2—CH_2—CH_2—CH_3$

9. Soda lime is–
(1) NaOH + CaO (2) NaOH + $Ca(OH)_2$
(3) Na_2CO_3 + CaO (4) $Na_2CO_3 + Ca(OH)_2$

10. $CH_3CH_2CH_2COOH \xrightarrow[\Delta]{\text{Soda lime}} 'A'$
$CH_3CH_2CH_2COOH \xrightarrow[\Delta]{\text{Red P + HI}} 'B'$
Product 'A' & 'B' are–
(1) Isomers (2) Identical
(3) Homologues (4) Conformer

11. Methane can prepared by–
(1) Kolbe's Electrolysis
(2) Wurtz reaction
(3) Carboxylic acid with soda lime
(4) Sabatier Senderens reaction

12. Kolbe's Electrolysis completed by–
(1) Ionic followed by free radical mechanism
(2) Free radical followed by ionic mechanism
(3) Only ionic mechanism
(4) Only free radical mechanism

13. Highest Boiling point is shown by–
(1) Pentane
(2) 2-Methyl butane
(3) 2,2-Dimethyl propane
(4) Butane

14. Iodination of methane can be carried out in presence of–
(1) HIO_3 (2) $HBrO_3$
(3) $HClO_3$ (4) HCl

15. During chlorination of methane which product is formed as a by product–
(1) $CH_3—CH_2—CH_3$ (2) $CH_3—CH_3$
(3) $CH_2(Cl)—CH_2(Cl)$ (4) $CH_3—CH_2—CH_2—CH_3$

16. Which of the following can be oxidised to alcohol by $KMnO_4$:–

(1) n-pentane (2) Iso pentane

(3) Neo pentane (4) propane

17. $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$ 'P' (major) Product 'P' is–

(1) CH_3OH (2) HCHO

(3) HCOOH (4) CH_3-CH_3

18. n-hexane $\xrightarrow[773K]{Cr_2O_3+Al_2O_3}$ Benzene

above reaction is known as–

(1) Aromatization (2) Reforming

(3) Cyclisation (4) All

19. Conversion of Higher alkane into lower alkene, alkane is called as–

(1) Isomerisation (2) Cracking

(3) Reforming (4) Oxidation

20. $CH_4 + H_2O \xrightarrow[1273K]{Ni}$ 'P'. Product 'P' is–

(1) Water gas (2) Producer gas

(3) Synthesis gas (4) Natural gas

21. Alkene is known as olefine due to–

(1) It is oily in nature

(2) It gives oily liqiuid on reaction with Cl_2

(3) It is obtained from oily liquid

(4) It give oil with water

22. How many sigma bond present in ethene have sp^2–sp^2 overlaping–

(1) 5 (2) 4

(3) 1 (4) 2

23. Correct reactivity order of Alkyl halide towards dehydrohalogenation–

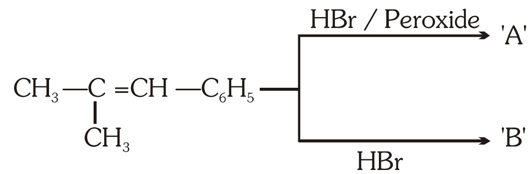
(1) 3° > 2° > 1° (2) 1° > 2° > 3°

(3) 2° > 1° > 3° (4) 1° > 3° > 2°

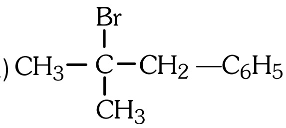
24. Acidic dehydration of alcohol is an example of–

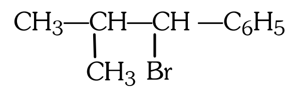
(1) α - Elimination (2) β- Elimination

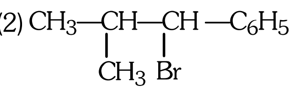
(3) γ- Elimination (4) None

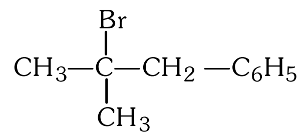
25. 

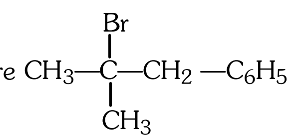
Product 'A' & 'B' are respectively–

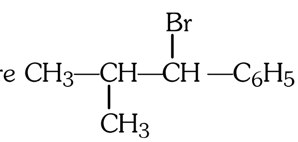
(1)  &

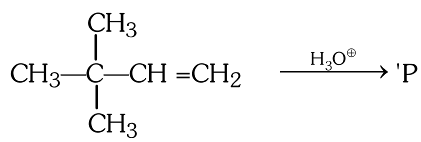


(2)  &

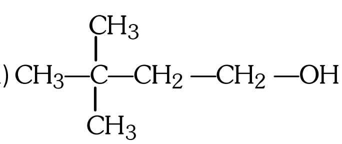


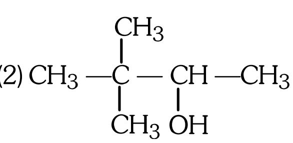
(3) Both are 

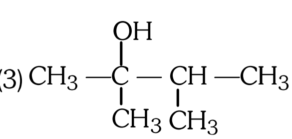
(4) Both are 

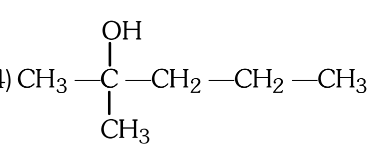
26.  (major)

product 'P' is–

(1) 

(2) 

(3) 

(4) 

27. Which of the following compound gives CO_2 on reductive ozonolysis–

(1) $CH_2 = CH - CH = CH_2$

(2) $CH_2 = C = CH - CH_3$

(3) $CH_3 - CH = CH - CH_3$

(4) $CH_2 = CH - CH_3$

28. Correct order of Acidic nature is–

(1) $CH \equiv CH > CH_3-C \equiv CH > CH_3-C \equiv C-CH_3$

(2) $CH_3-C \equiv C-CH_3 > CH_3-C \equiv CH > CH \equiv CH$

(3) $CH \equiv CH > CH_3-C \equiv C-CH_3 > CH_3-C \equiv CH$

(4) $CH \equiv CH < CH_3-C \equiv C-CH_3 < CH_3-C \equiv CH$

29. $$'x' + H_2O \xrightarrow[H_2SO_4]{HgSO_4} CH_3-\underset{\overset{\|}{O}}{C}-CH_2-CH_3$$

reactant 'x' in above reaction is–

(1) $CH_3-CH_2-C \equiv CH$

(2) $CH_3-C \equiv C-CH_3$

(3) $CH_3 - C \equiv CH$

(4) Both (1) & (2)

30. $CH \equiv CH + \underset{(Excess)}{NaNH_2} \longrightarrow$ 'P'

Final product 'P' is–

(1) $NaC \equiv CNa$ (2) $CH \equiv CNa$

(3) CH_3-CHO (4) No Reaction

31. Propanal and pentan -3-one are ozonolysis product of–

(1) $CH_3-CH_2-CH=\underset{\underset{CH_2-CH_3}{|}}{C}-CH_2-CH_3$

(2) $CH_3-CH=\underset{\underset{CH_2-CH_3}{|}}{C}-CH_2-CH_3$

(3) $\begin{array}{c} CH_3-CH_2-C-CH_3 \\ \| \\ CH_3-C-CH_3 \end{array}$

(4) $CH_3-CH=CH-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2-CH_3$

32. Each of the following can decolourise the reddish brown colour of $Br_2|CCl_4$, Except -

(1) C_2H_2 (2) C_2H_4

(3) Benzene (4) Both (1) & (3)

33. How many acyclic isomers of Alkyne C_6H_{10} are possible–

(1) 3 (2) 4 (3) 5 (4) 7

34. Which one is used in manufacture of non stick cookwares.

(1) Poly propene (2) Teflon

(3) Bakelite (4) Poly vinyl chloride

35. $CH_3 - CH = CH_2 + O_3 \longrightarrow$ 'P'

Product 'P' is–

(1) Ethanal + methanal

(2) Ethanoic acid + Methanoic acid

(3) Propene ozonide

(4) Acetone + Formaldehyde